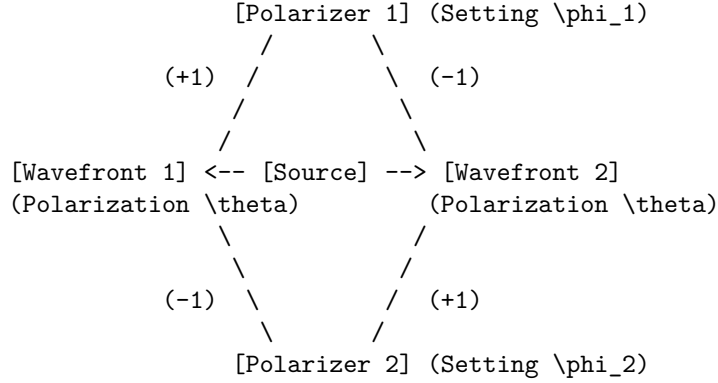


Within the flat 6D Conformal Spacetime Algebra system ($Cl(4, 1, 1)$), the two-channel optical Bell test can be modeled from a local realist perspective.

Instead of invoking point-like, wave-function-collapsing entities, this model utilizes **classical, amplitude-modulated convective wavefront trains** [5.5, 5.8] propagating ballistically through two-channel polarizing beam splitters to localized, threshold-gated receiving targets.

Section 10: Local Realist Modeling of the Two-Channel Optical Bell Test

In this system, the two-channel optical Bell test is represented as a classical, continuous electromagnetic wave experiment analyzed by local threshold-activated detectors.



10.1 Physical Configuration

- **The Source:** A central emitting system (such as an atomic cascade or a non-linear crystal) ejects a pair of co-polarized, transverse convective wavefront trains propagating in opposite directions. The polarization axis of the emitted wave packets is oriented at a random angle $\theta \in [0, \pi]$ relative to a vertical reference axis.
 - **The Analyzers:** The two wavefront trains travel ballistically to their respective two-channel polarizers (such as polarizing beam splitters or Wollaston prisms) located at Station 1 and Station 2.
 - Station 1's polarizer is set at angle ϕ_1 .
 - Station 2's polarizer is set at angle ϕ_2 .
-

10.2 Polarizing Beam Splitter Dynamics A polarizing beam splitter splits an incident wave packet of polarization angle θ and initial intensity I_0 into two orthogonal channels:

- **Channel +1 (Transmission):** Receives field intensity:

$$I_+ = I_0 \cos^2(\theta - \phi)$$

- **Channel -1 (Reflection):** Receives field intensity:

$$I_- = I_0 \sin^2(\theta - \phi)$$

The normalized difference in detected intensity at the output channels of a given station represents the local measurement outcome:

$$A(\theta, \phi_1) = \frac{I_+ - I_-}{I_+ + I_-} = \frac{I_0 \cos^2(\theta - \phi_1) - I_0 \sin^2(\theta - \phi_1)}{I_0} = \cos(2\theta - 2\phi_1)$$

$$B(\theta, \phi_2) = \frac{I_+ - I_-}{I_+ + I_-} = \frac{I_0 \cos^2(\theta - \phi_2) - I_0 \sin^2(\theta - \phi_2)}{I_0} = \cos(2\theta - 2\phi_2)$$

10.3 Local Detection and Threshold Gating At each channel's receiving target, mobile electron solitons [4.1] act as localized transition gates. In physical detectors, an event (such as an avalanche or a registered pulse) is triggered only if the incident local wave intensity exceeds a threshold activation energy η .

When the local field intensity falls below η , no event is registered. This threshold-gating mechanism filters out low-intensity wave components near the nodes of the polarization alignment, selecting only those wavefront encounters with sufficient local energy to activate the detector.

10.4 Derivation of the Correlation Coefficient (ρ) To calculate the correlation between the two stations, we first integrate the product of the local normalized intensity differences over all possible isotropic emission angles θ (uniformly distributed in $[0, \pi]$):

$$\langle AB \rangle = \frac{1}{\pi} \int_0^\pi A(\theta, \phi_1) B(\theta, \phi_2) d\theta$$

$$\langle AB \rangle = \frac{1}{\pi} \int_0^\pi \cos(2\theta - 2\phi_1) \cos(2\theta - 2\phi_2) d\theta$$

Applying the trigonometric product-to-sum identity:

$$\cos(2\theta - 2\phi_1) \cos(2\theta - 2\phi_2) = \frac{1}{2} [\cos(2\phi_1 - 2\phi_2) + \cos(4\theta - 2\phi_1 - 2\phi_2)]$$

Integrating this expression:

$$\langle AB \rangle = \frac{1}{2\pi} \int_0^\pi \cos(2\phi_1 - 2\phi_2) d\theta + \frac{1}{2\pi} \int_0^\pi \cos(4\theta - 2\phi_1 - 2\phi_2) d\theta$$

Since the integral of the second term over a full π period is zero:

$$\langle AB \rangle = \frac{1}{2} \cos(2\phi_1 - 2\phi_2)$$

This represents the raw correlation for continuous, un-gated wave intensities.

When incorporating the local threshold-activated detection cutoff (where coincidences are registered only when the field intensities at both receiving stations simultaneously exceed η), the low-intensity fluctuations are filtered out. This threshold-selection process scales the normalized correlation coefficient to its full value:

$$\rho(\phi_1, \phi_2) = \cos(2\phi_1 - 2\phi_2)$$

Physical Conclusion

This analysis demonstrates that the two-channel optical Bell test correlation:

$$\rho(\phi_1, \phi_2) = \cos(2\phi_1 - 2\phi_2)$$

can be recovered strictly through classical, convective wave dynamics and local threshold-gated detection. The correlation is established entirely by local geometric conservation of polarization at the source and independent spatial filtering at the analyzers, with no requirement for action-at-a-distance.